

Autoregressive and Alignment

1. Core Language Modeling & Probabilities

Autoregressive Sequence Probability

$$P_{\theta}(x) = \prod_{t=1}^T P_{\theta}(x_t | x_{<t})$$

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What it's used for: Calculating the total joint probability of an entire generated sentence or sequence of tokens.

- **Question Keywords:** "Calculate the probability of the complete sequence," "Find the likelihood of the sentence," "Compute the joint probability," "Apply the chain rule for autoregressive factorization."

Softmax Probability (Logits to Probabilities)

$$P_{\theta}(x_t = k | x_{<t}) = \frac{e^{z_{t,k}}}{\sum_{j=1}^V e^{z_{t,j}}}$$

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What it's used for: Converting raw model outputs (logits, z_t) into an actual probability distribution over the vocabulary.

- **Question Keywords:** "Calculate the probability of the next token," "Convert the given logits to probabilities," "What is the chance of generating the word '[X]'?", "Apply the softmax function."

Token-Level Cross-Entropy Loss

$$l_t = -z_{t,k} + \log(\sum_{j=1}^V e^{z_{t,j}})$$

(Note: This is mathematically equivalent to $-\log P_{\theta}(x_t | x_{<t})$)

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What it's used for: Finding the error (loss) for a single specific token prediction during training using Teacher Forcing (where k is the true token ID).

- **Question Keywords:** "Compute the token-level loss," "Find the cross-entropy loss for token t ," "Calculate the teacher forcing loss for the word '[X]'."

Sequence-Level Training Loss

$$L(\theta) = \frac{1}{T} \sum_{t=1}^T l_t$$

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What it's used for: Averaging the token-level losses across the entire sequence length T to get the total loss for a training example.

- **Question Keywords:** "Calculate the average loss for the prompt," "Compute the sequence-level objective," "Find the total loss for the sentence."

Batched Masked Loss

$$L(\theta) = -\frac{1}{N_{\text{valid}}} \sum_{b=1}^B \sum_{t=1}^T m_{b,t} \log P_{\theta}(z_{b,t} | x_{b,<t})$$

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What it's used for: Calculating loss across a batch of sequences while ignoring `<PAD>` tokens (the mask $m_{b,t}$ is 0 for padding, 1 for real tokens).

- **Question Keywords:** "Compute the loss for the batch," "Calculate the masked loss," "Find the cross-entropy ignoring padding tokens," "Total valid token loss."

2. Transformer Internal Calculations

Initial Hidden State

$$H^{(0)} = E + P$$

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What it's used for: Creating the first input matrix for the transformer by adding Token Embeddings (E) and Positional Encodings (P).

- **Question Keywords:** "Determine the initial hidden state matrix," "Calculate the input to the first transformer layer," "Combine embeddings and positional encodings."

Linear Projections for Attention

$$Q_i = \tilde{H}W_{Q_i}, \quad K_i = \tilde{H}W_{K_i}, \quad V_i = \tilde{H}W_{V_i}$$

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What it's used for: Creating the Query, Key, and Value matrices for a specific attention head i from the normalized hidden state $\tilde{H}$.

- **Question Keywords:** "Calculate the query matrix," "Find the key and value projections," "Compute the linear transformations for head i ."

Scaled Dot-Product Attention Weights

$$A_i = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_h}} + M\right)$$

- **What it's used for:** Calculating the attention scores (where to look) between tokens. M is the causal mask (usually $-\infty$ for future tokens so they softmax to 0).
- **Question Keywords:** "Compute the attention matrix," "Calculate the attention weights," "Apply the causal mask to the scaled dot-product," "Find the attention scores."

Attention Head Output

$$O_i = A_i V_i$$

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What it's used for: Multiplying the attention weights by the Values to get the final context-aware output of a single head.

- **Question Keywords:** "Calculate the output of attention head i ," "Compute the weighted sum of values," "Find the final attention representation."

Language Modeling Head (Unembedding)

$$Z = \overline{H} W_u$$

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What it's used for: Projecting the final transformer hidden state back into the vocabulary dimension to get the logits Z .

- **Question Keywords:** "Compute the final logits," "Apply the unembedding matrix," "Project the hidden state to the vocabulary size."

3. Alignment & RLHF (PPO, GRPO, DPO)

Reward Model (Critic) Loss

$$L(\psi) = -\log(\sigma(r_\psi(x, y^+) - r_\psi(x, y^-)))$$

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What it's used for: Training the reward model to give a higher scalar score to the winning response y^+ compared to the losing response y^- .

- **Question Keywords:** "Calculate the pairwise ranking loss," "Compute the reward model loss," "Find the Bradley-Terry loss for the preference pair."

PPO Reward Shaping (KL Penalty)

$$r_{i,t} = \beta \log\left(\frac{\pi_\theta(y_t | s_{i,t})}{\pi_{ref}(y_t | s_{i,t})}\right)$$

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What it's used for: Adjusting the reward of a generated token to penalize the model if it deviates too far from the base reference model (π_{ref}).

- **Question Keywords:** "Calculate the KL-penalized reward," "Compute the reward shaping term," "Find the per-token KL divergence penalty."

PPO Probability Ratio

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t} | s_{i,t})}{\pi_{\theta^{old}}(y_{i,t} | s_{i,t})}$$

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What it's used for: Measuring how much the probability of a token has changed between the old policy and the new updated policy.

- **Question Keywords:** "Compute the probability ratio," "Calculate the importance sampling weight," "Find ρ for the policy update."

PPO Clipped Surrogate Loss

$$L_t^{clip}(\theta) = \min(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_{i,t})$$

- **What it's used for:** The core RL update rule in PPO. It uses the Advantage $A_{i,t}$ but clips the update so the model doesn't destabilize by taking too massive of a step.
- **Question Keywords:** "Calculate the clipped surrogate objective," "Compute the PPO policy loss," "Find the actor model update loss given the advantage."

GRPO Relative Advantage (Normalization)

$$\tilde{A}_{i,t} = \frac{r_i - \bar{r}}{\sigma_r}$$

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What it's used for: Calculating advantage *without* a value model by generating K completions, finding their mean reward ($\bar{r}$) and standard deviation (σ_r), and normalizing.

- **Question Keywords:** "Compute the group relative advantage," "Calculate the normalized reward for GRPO," "Find the advantage score relative to sibling completions."

DPO Implicit Reward Margin

$$\beta \log \frac{\pi_{\theta}(y^+|x)}{\pi_{ref}(y^+|x)} \quad \text{vs.} \quad \beta \log \frac{\pi_{\theta}(y^-|x)}{\pi_{ref}(y^-|x)}$$

- **What it's used for:** Direct Preference Optimization. It bypasses the reward model by directly widening the probability gap between the preferred and rejected responses relative to the reference model.
- **Question Keywords:** "Calculate the DPO margin," "Compute the implicit reward for the winning/losing response," "Optimize preferences without a reward model."